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Pseudo-potential Lattice Boltzmann Method with an Improved Forcing Scheme for the Cumulant Collision Model

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Abstract
This paper proposes an improved cumulant collision model for the pseudo-potential lattice Boltzmann method (LBM) to increase the stability of multiphase flow simulations involving low viscosities. This model is based on the work of Kharmiani et al. in (J Stat Phys 175: 47, 2019), which can be extended regardless of the collision model. The original cumulant collision model (Geier et al. in Comput Math Appl 70:507, 2015) causes a non-physical shape of droplets in pseudo-potential LBM because only the first-order central moments are considered in the forcing scheme. The improved cumulant collision model proposed in this paper applies the central moment forcing scheme to the original cumulant model to cover the high-order central moments. Several numerical simulations were carried out to validate the proposed model. First, the problem of a stationary liquid layer was solved, where the proposed model was demonstrated to be thermodynamically consistent. Second, the problem of a stationary droplet was solved, where the result agreed well with Laplace's law. Third, the problem of a droplet impact on a liquid film was solved, where the crown radius agreed well with the analytical and numerical results available. Fourth, the simulation results carried out with the raw moment, central moment, and the proposed improved cumulant collision models were compared, as the liquid and vapor viscosities were gradually lowered. With all else being equal, only the lattice Boltzmann method with the proposed improved cumulant collision model was able to successfully simulate a density ratio of **720** and a Reynolds number of $\mathbf{8.7\times10^4}$.

Keywords Multiphase · Pseudo-potential LBM · Cumulant collision model · Droplet impact

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1 Introduction

About three decades ago, the lattice Boltzmann method (LBM) was developed as an alternative numerical scheme to simulate fluid flows [1]. Unlike conventional computational fluid dynamics methods such as the finite difference method, finite element method, and finite volume method, which are macroscopic approaches based on the Navier–Stokes equations, LBM has distinct features as a mesoscopic approach based on kinetic theory. These features include a simple collision-and-streaming-based algorithm, easy implementation of boundary conditions, and high parallel scalability [2]. Over the past three decades, LBM has been extensively studied in the fields of incompressible [3], compressible [4], thermal [5], multiphase [6], and multi-component [7] flows, with successful results.

In particular, the pseudo-potential LBM in the fields of multiphase, multi-component flows is attracting the attention of many researchers. Pseudo-potential LBM creates phase separation by applying an interaction force to the velocity field. Unlike other multiphase, multi-component LBM models such as the color-gradient model [8] and phase-field-based model [9], an additional effort for interface tracking is not required in the pseudo-potential LBM. Therefore, the pseudo-potential LBM allows multiphase, multi-component flow simulations to be efficient, with a lower memory overhead and a lower computational cost. However, the original pseudo-potential LBM also has several limitations, such as spurious velocities occurring near the liquid–vapor interface for problems with high density ratios and low viscosities that causes numerical instabilities; a non-physical phenomenon in which the liquid and vapor densities depend on the viscosities; violation of thermodynamic consistency; and dependency among the surface tension and the fluid densities. Researchers have continued to study of ways to improve said limitations.

Shan and Chen [10] first presented the idea of simulating multiphase, multi-component flows using the pseudo-potential LBM. Yuan and Schaefer [11] pointed out that Shan and Chen [10] used only the Shan-Chen (S-C) equation of state (EOS), and applied various EOS such as van der Waals, Redlich-Kwong, Redlich-Kwong Soave, Peng-Robinson, and Carnahan-Starling to reduce the spurious velocities near the interface. Kupershtokh et al. [12] pointed out that the S-C forcing scheme makes the fluid densities change according to the relaxation time, and suggested an exact difference method (EDM) forcing scheme as a solution. In addition, the forcing scheme used local and mean-value approximations to achieve thermodynamic consistency. Li et al. [13] analyzed the S-C forcing scheme and the EDM forcing scheme, and presented an improved forcing scheme based on Guo et al. [14]. Based on the local approximation forcing scheme, modified velocities were applied to achieve thermodynamic consistency. A detailed discussion of the forcing schemes, such as S-C, EDM, and Guo, which are important for the numerical stability and accuracy of the pseudo-potential LBM, can be found in review papers [15, 16].

However, due to the low viscosities, the spurious velocities near the interface increase, and instability occurs. With LBM, even when solving single phase flows, numerical instability occurs when the numerical viscosity decreases and the relaxation time approaches 0.5 [17]. Researchers have improved the collision model to increase the stability of the LBM. The stability of the LBM is improved by applying the raw moment (RM) collision model, which uses multiple relaxation times, instead of a single relaxation time [18]. When using multiple relaxation times, all relaxation times except those related to the kinematic viscosity are set as free parameters to increase the numerical stability. A central moment (CM) collision model which collides in a moving frame of reference was developed afterward to further increase the numerical stability [19]. Furthermore, a central Hermite moment (CHM) collision model

results in an expanded stability domain [20]. Geier et al. [3] presented a cumulant collision model. The cumulants are statistically independent, whereas dependencies exist between different moments in the RM and CM. The problem of flow around a sphere with low-viscosity fluids was successfully performed by applying the cumulant collision model.

Much of research in pseudo-potential LBM also aims to increase the numerical stability by improving the collision model. Yu et al. [21] applied the RM collision model to pseudo-potential LBM to reduce spurious velocities. Wu et al. [22] increased the stability by increasing the value of the relaxation time related to the bulk viscosity. Luo et al. [23] increased the stability by applying the CM collision model. Recently, Hosseini et al. [24] applied the CHM-based entropic collision model to the pseudo-potential LBM, enabling stable 3D droplet impact simulations.

Pseudo-potential LBM in previous studies was unable to independently modify the density ratio and surface tension. Sbragaglia et al. [25] used a multirange potential to change the surface tension. However, Huang et al. [26] pointed out that the density ratio is changed when the surface tension is changed in the model based on Sbragaglia et al. [25]. Li et al. [27] changed the surface tension independent of the density ratio by adding a source term to the RM collision model. Kharmiani et al. [6] presented an alternative method that can change the surface tension independent of the density ratio. The original forcing term was modified to satisfy thermodynamic consistency, and no additional source terms were applied in the collision model. Therefore, unlike the model based on Li et al. [27], the model based on Kharmiani et al. [6] has a high scalability regardless of the dimension, number of discrete velocities, and collision model. The stability was also improved by applying the RM collision model, as demonstrated in [6]. Recently, researchers validated that thermodynamic consistency is satisfied at very low temperatures with the density ratio exceeding 10^{10} [28–30]. Although many studies have greatly improved the numerical stability of the pseudo-potential LBM, limitations remain in simulating flows involving low viscosities.

In this paper, a cumulant collision model was applied to the model proposed by Kharmiani et al. [6] to increase the stability when simulating flows with low viscosities. To the best of the authors' knowledge, there are no studies to date where the cumulant collision model has been applied to the pseudo-potential LBM. When applying a forcing term in the cumulant model presented by Geier et al. [3], only the first-order central moments are considered, which causes a non-physical shape of the droplet, as seen in pseudo-potential LBM results of Sect. 3.2. In our improved cumulant collision model, the limitations of the original cumulant collision model are overcome by applying a central moment forcing scheme that considers higher-order central moments.

The rest of the paper is as follows. Section 2 introduces the pseudo-potential LBM with an improved cumulant collision model in two-dimensional space. In Sect. 3, numerical simulations are conducted to validate the proposed model and confirm the proposed model's stability. Sect. 4 draws a brief conclusion. Additionally, RM and CM collision models in two-dimensional space are presented in Appendix A and B, respectively. Finally, the improved cumulant collision model in three-dimensional space is presented in Appendix C.

2 Numerical Method

2.1 Original Cumulant Collision Model

The lattice Boltzmann equations with the Bhatnagar-Gross-Krook (BGK) collision operator including the forcing term are as follows [31, 32]:

$$f^{*}_{ijxyt} = \omega f^{eq}_{ijxyt} + (1-\omega) f_{ijxyt} + \Delta t(1-\omega/2) f^{F}_{ijxyt}, \tag{1}$$

$$f_{ij(x+ic\Delta t)(y+jc\Delta t)(t+\Delta t)} = f^{*}_{ijxyt}, \tag{2}$$

where f is the particle distribution function; (x, y) are the space variables; t is the time variable; ω is the relaxation rate; Δt is the time step; $c = \Delta x/\Delta t$; Δx is the grid spacing; $i, j \in \{1, 0, \bar{1}\}$ with $\bar{1}\equiv$-1; and f^{F} is the discrete forcing term. Also, * denotes the post-collision variables and ^{eq} denotes the equilibrium variables.

In this paper, the D2Q9 velocity set is used with the lattice units, $\Delta x = \Delta t = 1$, and lattice speed of sound $c_s = 1/\sqrt{3}$.

Geier et al. [3]'s efficient implementation of equilibrium particle distribution functions is adopted:

$$f^{eq}_{ij} = \rho X_i Y_j, \tag{3}$$

where ρ is the fluid density.

One-dimensional populations X_i, Y_j can be obtained from the one-dimensional moment system on the D1Q3 velocity set [33]. The one-dimensional Maxwellian equilibrium particle distribution function in ξ-space is as follows [17]:

$$f^{eq} = \rho\left(\frac{3}{2\pi}\right)^{1/2} exp\left\{-\frac{3}{2}(\xi-u)^2\right\} \tag{4}$$

The first three moments of the Maxwellian are obtained as follows:

$$M^{M}_{(0)} = \int_{-\infty}^{\infty} f^{eq} d\xi = \rho, \tag{5}$$

$$M^{M}_{(1)} = \int_{-\infty}^{\infty} \xi f^{eq} d\xi = \rho u, \tag{6}$$

$$M^{M}_{(2)} = \int_{-\infty}^{\infty} \xi^2 f^{eq} d\xi = \rho\left(u^2 + \frac{1}{3}\right). \tag{7}$$

Meanwhile, the discrete equilibrium particle distribution functions on the D1Q3 velocity set are as follows:

$$f^{eq}_{i} = \rho X_i. \tag{8}$$

The first three moments on the D1Q3 velocity set are obtained as follows:

$$M_{(0)} = \sum_{i=-1}^{1} f^{eq}_{i}, \tag{9}$$

$$M_{(1)} = \sum_{i=-1}^{1} i f^{eq}_{i}, \tag{10}$$

$$M_{(2)} = \sum_{i=-1}^{1} i^2 f^{eq}_{i}. \tag{11}$$

X_i are obtained from 3 by 3 linear system by combining Eqs. (5)-(7) and Eqs. (9)-(11). In the same way, Y_j can also be obtained as follows:

$$X_{\bar{1}} = (u^2 - u + 1/3)/2, \tag{12}$$
$$X_0 = 2/3 - u^2, \tag{13}$$
$$X_1 = (u^2 + u + 1/3)/2, \tag{14}$$
$$Y_{\bar{1}} = (v^2 - v + 1/3)/2. \tag{15}$$
$$Y_0 = 2/3 - v^2, \tag{16}$$
$$Y_1 = (v^2 + v + 1/3)/2, \tag{17}$$

where (u, v) are the fluid velocities for the (x, y) directions, respectively.

Geier et al. [3] transformed the particle distribution functions from the velocity space to the cumulant space, and performed collision in the cumulant space to increase the stability. The cumulants, $c_{\alpha\beta}$, are defined by [3] as follows:

$$c_{\alpha\beta} = \frac{\partial^\alpha \partial^\beta}{\partial \Xi^\alpha \partial \Upsilon^\beta} \ln \left(\sum_{i,j} f_{ij} e^{\Xi i} e^{\Upsilon j} \right) \Bigg|_{\Xi=\Upsilon=0}, \tag{18}$$

where Ξ and Υ denote the velocity space wavenumbers, and $\alpha, \beta \in \{0, 1, 2\}$.

Since the direct transformation from the velocity space to the cumulant space is computationally expensive, the cumulants, $c_{\alpha\beta}$, are computed through the central moments, $k_{\alpha\beta}$, as defined by [3]

$$\begin{aligned} k_{\alpha\beta} &= \frac{\partial^\alpha \partial^\beta}{\partial \Xi^\alpha \partial \Upsilon^\beta} \left(e^{-\Xi u} e^{-\Upsilon v} \sum_{i,j} f_{ij} e^{\Xi i} e^{\Upsilon j} \right) \Bigg|_{\Xi=\Upsilon=0} \\ &= \sum_{i,j} (i-u)^\alpha (j-v)^\beta f_{ij}. \end{aligned} \tag{19}$$

Computation of the central moments can be simply split into two directions using a forward central moment transformation [3]:

$$k_{i|\beta} = \sum_j f_{ij} (j - v)^\beta, \tag{20}$$
$$k_{\alpha\beta} = \sum_i k_{i|\beta} (i - u)^\alpha. \tag{21}$$

The forward cumulant transformation can be derived from Eqs. (18)-(19):

$$C_{20} = k_{20}, \tag{22}$$
$$C_{02} = k_{02}, \tag{23}$$
$$C_{11} = k_{11}, \tag{24}$$
$$C_{12} = k_{12}, \tag{25}$$
$$C_{21} = k_{21}, \tag{26}$$
$$C_{22} = k_{22} - \frac{k_{20}k_{02} + 2k_{11}^2}{\rho}. \tag{27}$$

In Eqs. (22)-(27), normalization is performed for convenience as $C_{\alpha\beta} = \rho c_{\alpha\beta}$. The rest of the cumulants such as (C_{00}, C_{10}, C_{01}) and central moments such as (k_{00}, k_{10}, k_{01}) can be

obtained from macroscopic values to conserve mass and momentum. Thus, the post-collision central moments such as (k^*_{00}, k^*_{10}, k^*_{01}) will be shown before the backward central moment transformation.

Collision in the cumulant space occurs as follows:

$$C^*_{20} - C^*_{02} = (1-\omega_1)(C_{20} - C_{02}), \tag{28}$$

$$C^*_{20} + C^*_{02} = \omega_2 \frac{2}{3}\rho + (1-\omega_2)(C_{20} + C_{02}), \tag{29}$$

$$C^*_{11} = (1-\omega_1)C_{11}, \tag{30}$$

$$C^*_{12} = (1-\omega_3)C_{12}, \tag{31}$$

$$C^*_{21} = (1-\omega_3)C_{21}, \tag{32}$$

$$C^*_{22} = (1-\omega_4)C_{22}, \tag{33}$$

where ω_1 is related to the kinematic viscosities of the fluids, ω_2 is related to the bulk viscosity, and (ω_3, ω_4) are free parameters with a value between 0 and 2. In Eqs. (28)-(30), ω_1 and ω_2 can be obtained by [22]

$$\omega_1 = \left(3\nu + \frac{1}{2}\right)^{-1}, \tag{34}$$

$$\omega_2 = \left(3\xi + \frac{1}{2}\right)^{-1}, \tag{35}$$

with

$$\rho\xi = \mu' + \frac{\rho\nu}{3}, \tag{36}$$

where ν is the kinematic viscosity. Wu et al. [22] mentioned that ξ is not the bulk viscosity, but rather μ' is the bulk viscosity, which can be derived from the Chapman-Enskog expansion [17]. Here ν can be interpolated using the kinematic viscosities of the liquid and the vapor as follows:

$$\nu = \nu_V + \frac{\rho - \rho_V}{\rho_L - \rho_V}(\nu_L - \nu_V), \tag{37}$$

where L and V denote liquid and vapor, respectively.

After the collision process, post-collision cumulants are transformed back to the central moments. Applying the post-collision operator to Eqs. (22)-(27), Eqs. (38)-(43) are obtained via the backward cumulant transformation:

$$k^*_{20} = C^*_{20}, \tag{38}$$

$$k^*_{02} = C^*_{02}, \tag{39}$$

$$k^*_{11} = C^*_{11}, \tag{40}$$

$$k^*_{12} = C^*_{12}, \tag{41}$$

$$k^*_{21} = C^*_{21}, \tag{42}$$

$$k^*_{22} = C^*_{22} + \frac{k^*_{20}k^*_{02} + 2k^{*2}_{11}}{\rho}. \tag{43}$$

2.2 Improved Forcing Scheme for the Cumulant Collision Model

Only the first-order central moments are considered in [3] when applying the forcing scheme.

Original forcing scheme for the cumulant collision model:

$$k_{10}^{*} = -k_{10}, \tag{44}$$

$$k_{01}^{*} = -k_{01}. \tag{45}$$

Geier et al. [3] mentioned that high-order central moments have no contribution to the forcing scheme because the particle distribution functions have been transformed into the moving frame of reference. As observed in the stationary droplet simulation results of Sect. 3.2, however, only considering the first-order central moments is insufficient to reflect the interaction forces in pseudo-potential LBM, so high-order central moments must be considered. To solve this problem, a central moment forcing scheme is used in the cumulant collision model after backward cumulant transformation as follows:

$$k_{\alpha\beta}^{F} = \sum_{i,j} (i-u)^{\alpha} (j-v)^{\beta} f_{ij}^{F}, \tag{46}$$

with

$$\begin{aligned} f_{ij}^{F} = \omega_{ij} \Bigg[& 3(iF_x + jF_y) \\ & + \frac{9}{2}\left\{\left(i^2 - \frac{1}{3}\right)2uF_x + \left(j^2 - \frac{1}{3}\right)2vF_y + 2ij(uF_y + vF_x)\right\} \\ & + \frac{27}{2}\left\{\left(i^2 - \frac{1}{3}\right)j(2uvF_x + u^2F_y) + i\left(j^2 - \frac{1}{3}\right)(2uvF_y + v^2F_x)\right\} \\ & + \frac{81}{4}\left\{\left(i^2 - \frac{1}{3}\right)\left(j^2 - \frac{1}{3}\right)(2u^2vF_y + 2uv^2F_x)\right\}\Bigg], \end{aligned} \tag{47}$$

where w_{ij} are the weight given by

$$w_{ij} = \frac{4}{9}\frac{1}{4^{|i|+|j|}}, \tag{48}$$

and $k_{\alpha\beta}^{F}$ are the discrete central moment forcing terms, and (F_x, F_y) are forces in the (x, y) directions, respectively.

In Eq. (47), the forcing scheme in Huang et al. [34], which considers high-order discrete forcing terms, is applied to our model. The popular forcing scheme in Guo et al. [14] is recovered from the forcing scheme in Huang et al. [34] if terms related to third and fourth-order of Hermite polynomials in Eq. (47) are neglected. De Rosis et al. [35] transformed the forcing scheme in Huang et al. [34] to the central moment space and confirmed that the central moment forcing terms, $k_{\alpha\beta}^{F}$, lead to velocity-independent forcing terms. In Eq. (47), we use high-order discrete forcing terms, considering only the Hermite polynomial tensors supported by standard velocity sets such as D2Q9 and D3Q27. The cubic error terms of the velocity due to a lack of symmetry of the third-order moments tensor can be covered by correction terms [34]. However, since this paper aims for the incompressible limit, correction terms have not been considered.

Cumbersome calculations with Eqs. (46)–(48) give the following expressions:

$$k_{10}^{F} = F_x, \tag{49}$$

$$k_{01}^{F} = F_y, \tag{50}$$

$$k_{12}^{F} = \frac{F_x}{3}, \tag{51}$$

$$k_{21}^{F} = \frac{F_y}{3}, \tag{52}$$

where all other values of $k^F_{\alpha\beta}$ are zero.

Equations (49)-(52) lend the improved cumulant forcing scheme, as follows:

$$\tilde{k}^*_{00} = \rho, \tag{53}$$

$$\tilde{k}^*_{10} = \frac{F_x}{2}, \tag{54}$$

$$\tilde{k}^*_{01} = \frac{F_y}{2}, \tag{55}$$

$$\tilde{k}^*_{20} - \tilde{k}^*_{02} = k^*_{20} - k^*_{02}, \tag{56}$$

$$\tilde{k}^*_{20} + \tilde{k}^*_{02} = k^*_{20} + k^*_{02}, \tag{57}$$

$$\tilde{k}^*_{11} = k^*_{11}, \tag{58}$$

$$\tilde{k}^*_{12} = k^*_{12} + \left(1 - \frac{\omega_3}{2}\right)\frac{F_x}{3}, \tag{59}$$

$$\tilde{k}^*_{21} = k^*_{21} + \left(1 - \frac{\omega_3}{2}\right)\frac{F_y}{3}, \tag{60}$$

$$\tilde{k}^*_{22} = k^*_{22}, \tag{61}$$

where $\tilde{k}^*$ denotes the post-collision central moment with the improved cumulant forcing scheme.

The post-collision central moments are transformed back to the velocity space using a splitting method similar to that in Geier et al. [3].

The detailed backward central moment transformation is described in Eqs. (62)-(67):

$$\tilde{k}^*_{0|\beta} = \tilde{k}^*_{0\beta}(1-u^2) - 2u\tilde{k}^*_{1\beta} - \tilde{k}^*_{2\beta}, \tag{62}$$

$$\tilde{k}^*_{\bar{1}|\beta} = \{\tilde{k}^*_{0\beta}(u^2-u) + \tilde{k}^*_{1\beta}(2u-1) + \tilde{k}^*_{2\beta}\}/2, \tag{63}$$

$$\tilde{k}^*_{1|\beta} = \{\tilde{k}^*_{0\beta}(u^2+u) + \tilde{k}^*_{1\beta}(2u+1) + \tilde{k}^*_{2\beta}\}/2, \tag{64}$$

$$f^*_{i0} = \tilde{k}^*_{i|0}(1-v^2) - 2v\tilde{k}^*_{i|1} - \tilde{k}^*_{i|2}, \tag{65}$$

$$f^*_{i\bar{1}} = \{\tilde{k}^*_{i|0}(v^2-v) + \tilde{k}^*_{i|1}(2v-1) + \tilde{k}^*_{i|2}\}/2, \tag{66}$$

$$f^*_{i1} = \{\tilde{k}^*_{i|0}(v^2+v) + \tilde{k}^*_{i|1}(2v+1) + \tilde{k}^*_{i|2}\}/2. \tag{67}$$

After the processes of collision, streaming, and boundary condition imposition are applied to the particle distribution functions, the macroscopic values are calculated as follows:

$$\rho = \sum_{i,j} f_{ij}, \tag{68}$$

$$\rho u = \sum_{i,j} i f_{ij} + \frac{F_x}{2}, \tag{69}$$

$$\rho v = \sum_{i,j} j f_{ij} + \frac{F_y}{2}. \tag{70}$$

Kharmiani et al. [6] proposed forcing terms that enable the surface tension to be modified independently of the density ratio, ρ_L/ρ_V, while satisfying thermodynamic consistency, as follows:

$$\boldsymbol{F} = \psi\nabla\psi - \frac{\lambda_0+\kappa}{12}\nabla\left(|\nabla\psi|^2\right) + \frac{\kappa}{6}\nabla^2\psi\nabla\psi, \tag{71}$$

with

$$\boldsymbol{F} = (F_x, F_y), \tag{72}$$

$$\nabla\psi = \left(\frac{\partial\psi}{\partial x}, \frac{\partial\psi}{\partial y}\right), \tag{73}$$

$$|\nabla\psi|^2 = \left(\frac{\partial\psi}{\partial x}\right)^2 + \left(\frac{\partial\psi}{\partial y}\right)^2, \tag{74}$$

$$\nabla^2\psi = \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2}. \tag{75}$$

The isotropic central schemes are adopted to calculate the gradients in Eqs. (73)-(74) and the Laplacian operator in Eq. (75) [36]:

$$\frac{\partial\psi}{\partial x} = \sum_{i,j} 3w_{ij} i \psi_{(x+i)(y+j)t}, \tag{76}$$

$$\frac{\partial\psi}{\partial y} = \sum_{i,j} 3w_{ij} j \psi_{(x+i)(y+j)t}, \tag{77}$$

$$\nabla^2\psi = \sum_{i,j} 6w_{ij}\{\psi_{(x+i)(y+j)t} - \psi_{xyt}\}, \tag{78}$$

where ψ is the interaction potential defined by [37]

$$\psi = \sqrt{-2\left(P_{EOS} - \frac{\rho}{3}\right)}. \tag{79}$$

In Eq. (71), λ_0 is adjusted to satisfy thermodynamic consistency, and κ is adjusted to change the surface tension.

In this paper, Carnahan-Starling (C-S) EOS is adopted as follows [11]:

$$P_{EOS} = \rho R_s T \frac{1 + b\rho/4 + (b\rho/4)^2 - (b\rho/4)^3}{(1 - b\rho/4)^3} - a\rho^2, \tag{80}$$

with

$$a = 0.4963\frac{R_s^2 T_c^2}{P_c}, \tag{81}$$

$$b = 0.18727\frac{R_s T_c}{P_c}, \tag{82}$$

where R_s is the specific gas constant, P_c is the critical pressure, and T_c is the critical temperature.

3 Results and Discussion

In this section, the target problem of droplet impact on a liquid film will be conducted to compare the numerical stability of the RM, CM, and the proposed cumulant models. Firstly, the problem of the stationary liquid layer is considered in Sect. 3.1 to obtain the initial density of the target problem. Secondly, the surface tension acting between the liquid and the gas is obtained to calculate the Weber number of the target problem. In pseudo-potential LBM, the surface tension can be obtained through the problem of stationary droplets as shown in

Table 1 The coexistence table for C-S EOS with $\lambda_0 = 2.65$

T/T_c	a	ρ_L/ρ_V	
		Improved cumulant	Maxwell construction
0.95	1.000	0.210/0.0668	0.210/0.0666
0.90	1.000	0.248/0.0457	0.248/0.0454
0.85	1.000	0.279/0.0318	0.279/0.0316
0.80	0.500	0.307/0.0211	0.307/0.0217
0.75	0.500	0.333/0.0140	0.333/0.0145
0.70	0.500	0.358/0.00916	0.358/0.00929
0.65	0.400	0.382/0.00560	0.382/0.00559
0.60	0.250	0.406/0.00286	0.406/0.00308
0.55	0.220	0.430/0.00152	0.430/0.00151
0.50	0.170	0.455/0.000631	0.454/0.000626
0.45	0.130	0.479/0.000193	0.479/0.000205
0.40	0.107	0.504/4.45E-05	0.504/4.41E-05
0.35	0.091	0.531/6.46E-06	0.531/6.43E-06
0.30	0.078	0.559/3.93E-07	0.559/3.79E-07

Sect. 3.2. In Sect. 3.3, the values from Sects. 3.1 and 3.2 are applied to the problem of droplet impact on a liquid film. This paper did not cover the CHM model, which is known for its high stability [20]. Further research on the CHM model will be conducted in future work.

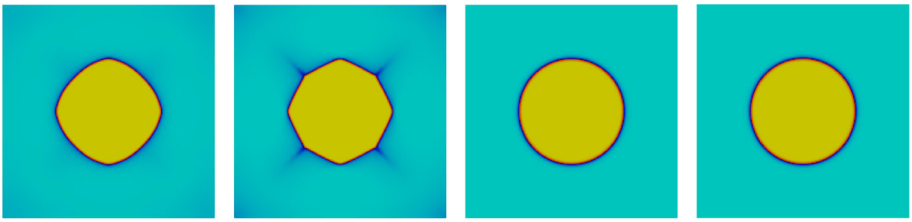
3.1 Stationary Liquid Layer

The problem of a stationary liquid layer was conducted to confirm that the improved cumulant model satisfies thermodynamic consistency. A liquid layer of thickness t is initially placed in the middle of a $N \times N$ domain bounded by 4 periodic boundary conditions. The initial density in space is applied as follows:

$$\rho = \rho_V + \frac{\rho_L - \rho_V}{2}\left[\tanh\left\{\frac{2}{W}\left(y - \frac{N-t}{2}\right)\right\} - \tanh\left\{\frac{2}{W}\left(y - \frac{N+t}{2}\right)\right\}\right], \tag{83}$$

where $N = 200$, $t = 50$, and the interface width $W = 5$. Initial values of ρ_L and ρ_V should be set near the values from Maxwell construction for numerical stability. ω_1 is set to 1.67 for the problem. Since the larger the bulk viscosity, the higher the LBM stability [22], ω_2 is set to 0.51, same as that in [6]. When ω_3 and ω_4 are set to 1, post-collision cumulants in Eqs. (31)-(33) become 0. Therefore, the computational cost can be reduced. In C-S EOS, parameters are set to $b = 4$, and $R_s = 1$, as were done in [11]. The interface width can be increased by reducing the parameter a in C-S EOS. Since surface tension effects are outside the scope of this paper, κ is set to 0.

Equilibrium density of the liquid and gas depending on the temperature are shown in Table 1. In this simulation, the value of λ_0 is set to 2.65, as was done in [6]. The coexistence table for C-S EOS matched well with the result of Maxwell construction. Since the pseudo-potential LBM is a diffuse-interface method, the width of the interface should be about 4 to 6 to ensure the accuracy and stability. As the temperature decreases, the width of the interface between



(a) (b) (c) (d)

Fig. 1 Stationary droplets at equilibrium state using (a) $\omega_1 = 1.67$ with the original cumulant model, Eqs. (44)-(45), (b) $\omega_1 = 1.0$ with the original cumulant model, Eqs. (44)-(45), (c) $\omega_1 = 1.67$ with the improved cumulant model, Eqs. (53)-(61), and (d) $\omega_1 = 1.0$ with the improved cumulant model, Eqs. (53)-(61)

the liquid and gas decreases. On the other hand, as the value of a decreases, the interface width increases, as discussed in [38]. Therefore, as the temperature decreases, the value of a increases, making the interface width about 4 to 6.

The results of the stationary liquid layer using $\lambda_0 = 2.65$ and $a = 0.17$ at $T = 0.5T_c$ are $\rho_L = 0.455$ and $\rho_V = 0.000631$, which agree well with the results of [6].

3.2 Stationary Droplet

The problem of the stationary droplet was conducted to obtain the surface tension between the liquid and vapor. A liquid droplet of radius R was placed in the middle of a $N \times N$ domain bounded by 4 periodic boundary conditions. The initial density in space is applied as follows:

$$\rho = \frac{\rho_L + \rho_V}{2} + \frac{\rho_L - \rho_V}{2} \tanh\left[\frac{2}{W}\left\{R - \sqrt{\left(x - \frac{N}{2}\right)^2 + \left(y - \frac{N}{2}\right)^2}\right\}\right], \tag{84}$$

where $N = 200$ and the other variables are set as the same as those in Sect. 3.1 unless otherwise specified.

When the original cumulant model is used, the droplets are quadrangular or octagonal, as shown in Fig. 1a, b. A non-physical phenomenon occurs in which the shape of droplets changes according to the ω_1. On the other hand, droplets become circular in the equilibrium state regardless of the ω_1 when the improved cumulant model is applied as shown in Fig. 1c, d. From these results, it is evident that the original cumulant forcing scheme is insufficient to reflect the interaction force in pseudo-potential LBM, and higher-order central moments must be considered.

In the results using the improved cumulant model, the pressure difference inside and outside the droplet was calculated by varying the radius R of the droplet. The result in Fig. 2 agrees well with Laplace's law which states that ΔP and R are inversely proportional. The value of the surface tension, as represented by the slope in the graph, is 0.0058, which is the same as the result of [6].

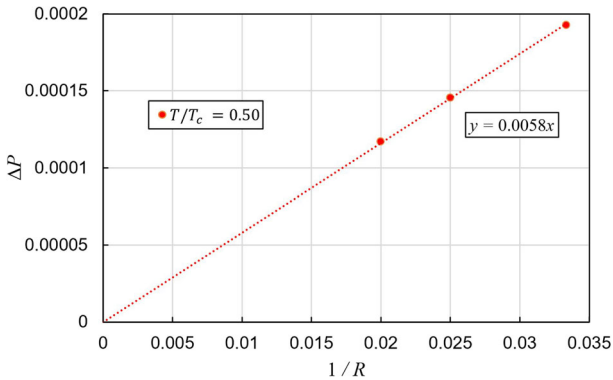


Fig. 2 Validation of Laplace′s law for a stationary droplet at $T = 0.5T_c$

3.3 Droplet Impact on a Liquid Film

The problem of a droplet impact on a liquid film is conducted to verify the stability of the improved cumulant model. A liquid droplet of radius R and a liquid film of thickness h were placed in a $2N \times N$ domain with periodic boundary conditions on the left and right. The top and bottom are set as fixed walls and the halfway bounce-back boundary conditions are applied to satisfy mass conservation [39]. The initial density in space is applied as follows:

$$\rho = \rho_L + \frac{\rho_L - \rho_V}{2} \left[\tanh\left\{ \frac{2}{W}\left(R - \sqrt{\left(x - \frac{N}{2}\right)^2 + (y - (R + h + W))^2} \right) \right\} + \tanh\left\{ \frac{2}{W}(h - y) \right\} \right], \tag{85}$$

with $N = 300$, $R = 40$, and $h = 20$.

The droplet collides into the liquid film with the initial speed V_0, and the initial velocities in space are as follows:

$$u = 0, \tag{86}$$

$$v = -\frac{V_0}{2} - \frac{V_0}{2} \tanh\left\{ \frac{2}{W}\left(R - \sqrt{\left(x - \frac{N}{2}\right)^2 - (y - (R + h + W))^2} \right) \right\}, \tag{87}$$

where $V_0 = 0.12$ and the other variables are set as the same as those in Sect. 3.1, unless otherwise specified.

The results of droplet impact on a liquid film at a Reynolds number $Re = \frac{2V_0 R}{\nu_L} = 1440$, Weber number $We = \frac{2\rho_L V_0^2 R}{\sigma} = 90$, and $\nu_V/\nu_L = 5$ are shown in Fig. 3. Our results in Fig. 4 agree well with the analytical solution derived by Yarin and Weiss [40] and the numerical solution of Kharmiani et al. [6].

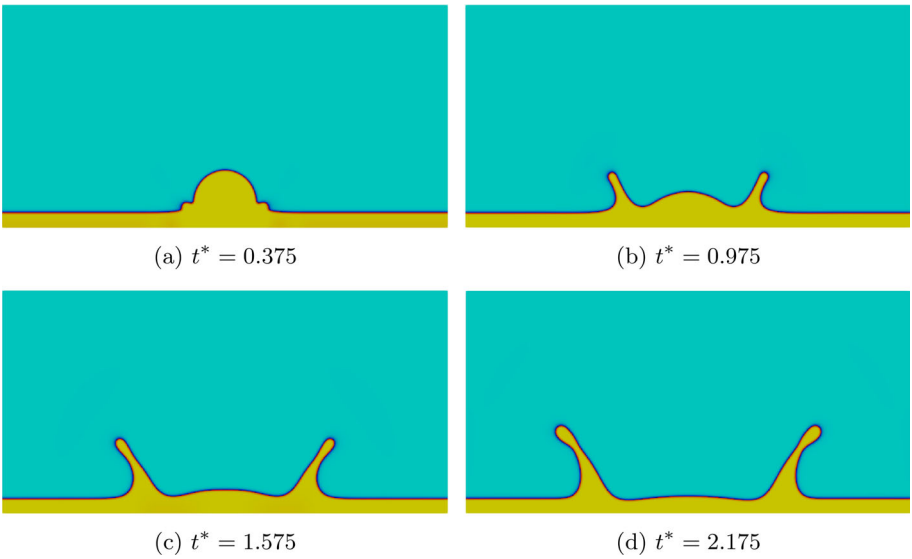


Fig. 3 Density contours during a droplet impact on a liquid film at $T = 0.5T_c$, $Re = 1440$, $We = 90$, and $\nu_V/\nu_L = 5$

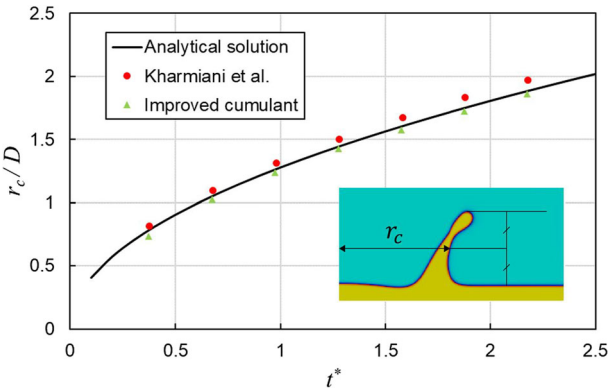


Fig. 4 Validation of a droplet impact on a liquid film compared with the analytical solution derived by Yarin and Weiss [40] and the numerial solution of Kharmiani et al. [6]

The analytical solution derived by Yarin and Weiss [40] is as follows:

$$\frac{r_c}{D} = \left(\frac{2D}{3h}\right)^{\frac{1}{4}} (t^*)^{\frac{1}{2}}, \tag{88}$$

where r_c is the radius of the crown, $D = 2R$, and $t^* = \frac{V_0}{D}t$.

The previous simulation was performed again with the RM and CM collision models to compare with the results of the improved cumulant model, with regard to the stability. The simulations of the RM, CM, and improved cumulant models produced the same results at $t^* = 2.175$, as seen in Fig. 5a–c. In this paper, the RM and CM collision models are used as shown in Appendix A and Appendix B , respectively. Other simulations were performed

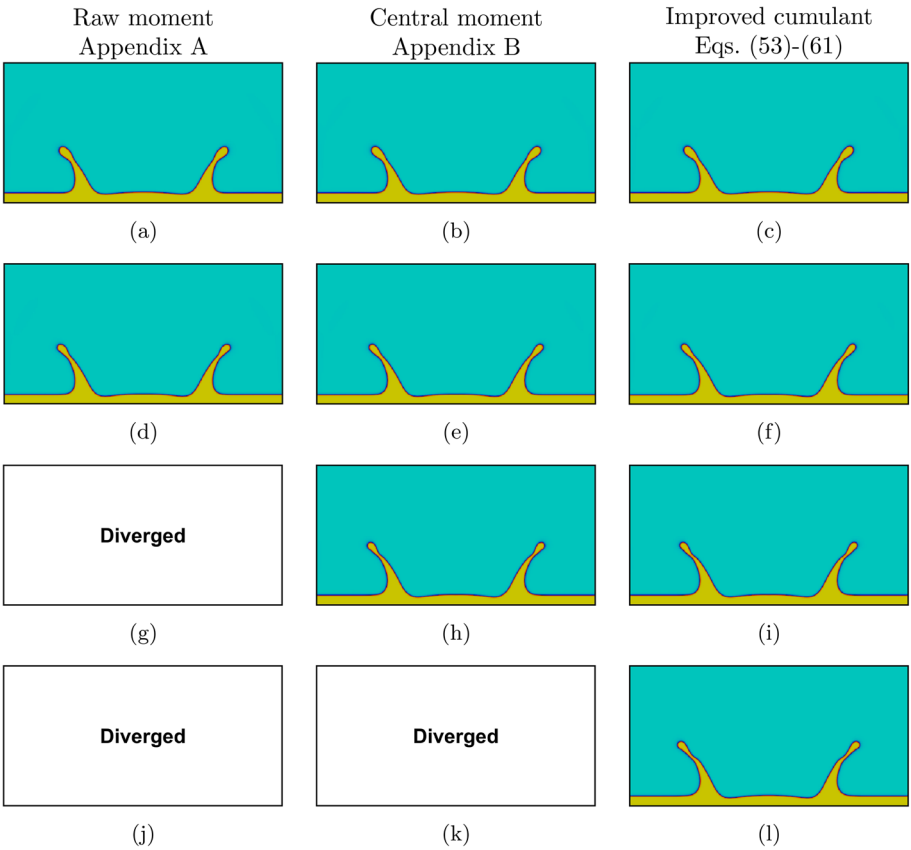


Fig. 5 Density contours during a droplet impact on a liquid film at $t^* = 2.175$ using (a, b, c) $\nu_L = 0.00667$, $\omega_2 = 0.51$, (d, e, f) $\nu_L = 0.002$, $\omega_2 = 0.46$, (g, h, i) $\nu_L = 0.0008$, $\omega_2 = 0.42$, and (j, k, l) $\nu_L = 0.00011$, $\omega_2 = 0.21$. Also, (a, d, g, j), (b, e, h, k), and (c, f, i, l) are obtained by the RM (Appendix A), CM(Appendix B), and improved cumulant (Eqs. (53)-(61)) collision models, respectively

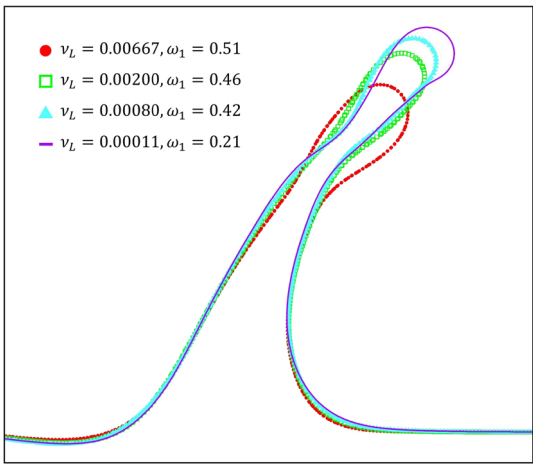


Fig. 6 Density contours during a droplet impact on a liquid film at $t^* = 2.175$ using (red circle) $\nu_L = 0.00667$, $\omega_2 = 0.51$, (green square) $\nu_L = 0.002$, $\omega_2 = 0.46$, (blue triangle) $\nu_L = 0.0008$, $\omega_2 = 0.42$, and (purple line) $\nu_L = 0.00011$, $\omega_2 = 0.21$ (Color figure online)

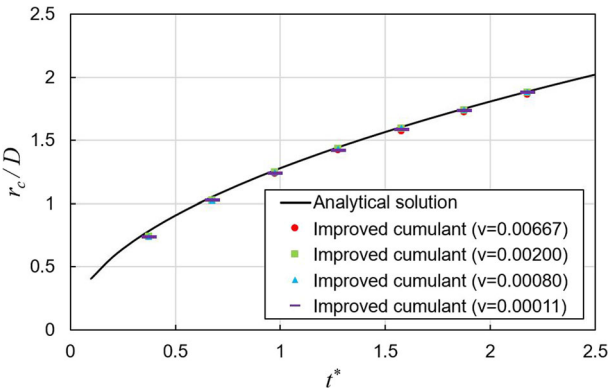


Fig. 7 Validation of a droplet impact on a liquid film compared with the analytical solution of [40] for various kinematic viscosities

Table 2 Comparison of the stability for the droplet impact on a liquid film according to ν_L and ω_2

ν_L	ω_2	RM	CM	Improved cumulant
0.00667	0.510	**stable**	**stable**	**stable**
0.00200	0.510	unstable	**stable**	**stable**
	0.485	unstable	**stable**	**stable**
	0.460	**stable**	**stable**	**stable**
0.00080	0.460	unstable	unstable	unstable
	0.440	unstable	unstable	**stable**
	0.420	unstable	**stable**	**stable**
0.00030	0.420	unstable	unstable	unstable
	0.385	unstable	unstable	unstable
	0.350	unstable	unstable	**stable**
0.00011	0.350	unstable	unstable	unstable
	0.280	unstable	unstable	unstable
	0.210	unstable	unstable	**stable**

reducing the kinematic viscosities while maintaining the same kinematic viscosity ratio. The simulations of a droplet impact on a liquid film are critically stable at $\nu_L = 0.002$ and $\omega_2 = 0.46$ for the RM model in Fig. 5d, and $\nu_L = 0.0008$ and $\omega_2 = 0.42$ for the CM model in Fig. 5h. On the other hand, the improved cumulant model was critically able to simulate at $\nu_L = 0.00011$ and $\omega_2 = 0.21$, corresponding to $Re = 8.7 \times 10^4$ for the problem of droplet impact shown in Fig. 5l. In Fig. 6, as the viscosities are decreased, the shear stress is reduced and the secondary droplets become thinner and more stretched. Furthermore, the results in Fig. 7 agree well with the analytical solution of [40] derived from mass and momentum equations, showing independence from kinematic viscosity.

Table 2 shows the stability of the results for the collision models for different ν_L and ω_2. The results of the RM, CM, and improved cumulant are all stable with ν_L=0.00667 and ω_2=0.51. When the value of ν_L was reduced to 0.002, only the result of the RM became unstable. At this time, the RM model can be solved stably by reducing ω_2 to 0.46. When the value of the bulk viscosity is increased, the stability of the LBM is improved in our simulations, as discussed in [22]. The result of the CM also became unstable with the reduced

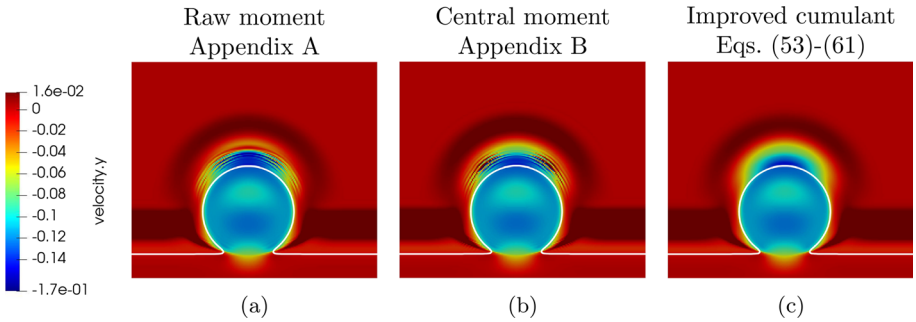


Fig. 8 Contours of the fluid velocity for the y direction at $t^* = 0.09$ using $\nu_L = 0.00011$, $\omega_2 = 0.21$. Also, (a), (b), and (c) are obtained by the RM (Appendix A), CM (Appendix B), and improved cumulant (Eqs. (53)-(61)) collision models, respectively

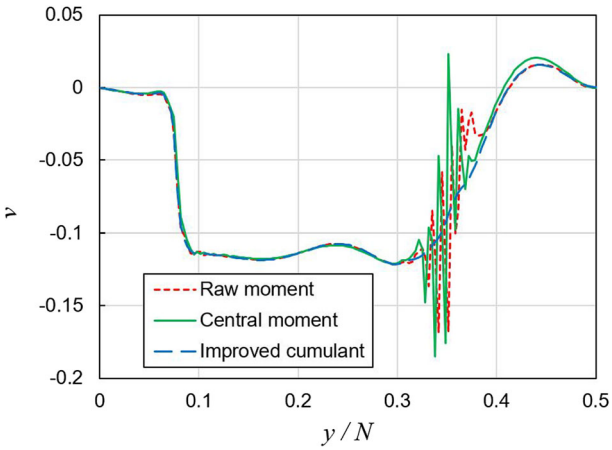


Fig. 9 The graph of the fluid velocity for the y direction at $x = (N + \frac{R}{2})$ and $t^* = 0.09$ using $\nu_L = 0.00011$, $\omega_2 = 0.21$

value of ν_L=0.0008 with ω_2=0.46, and the stability can be recovered by reducing ω_2 to 0.42. However, when the value of ν_L became smaller than 0.0008, the stability of the result of the CM could not be secured even when ω_2 was reduced. On the other hand, our improved cumulant model can be utilized stably even when ν_L was 0.00011 by reducing ω_2 to 0.21.

Contours and the graph of the fluid velocity for the y direction at $t^* = 0.09$ using $\nu_L = 0.00011$, $\omega_2 = 0.21$ are shown in Figs. 8 and 9, respectively. In the results of the RM and CM models, oscillations that cause instability are observed near the top of the droplet due to low viscosities, whereas the result of our improved cumulant model showed a smooth velocity contour. These simulations confirmed that the improved cumulant model is more stable than the RM and CM models at low kinematic viscosities.

4 Conclusion

In this paper, a pseudo-potential LBM with a cumulant collision model is presented for solving problems involving low viscosities. The problem of droplet impact on a liquid film

was adopted as the target problem to compare the stability of the improved cumulant model with other models such as the RM and CM. Firstly, the problem of a stationary liquid layer was performed to obtain the densities of the liquid and vapor to be used in the target problem. Also, we confirmed that the improved cumulant LBM is thermodynamically consistent. Secondly, the problem of a stationary droplet was performed to obtain the surface tension to be used in the target problem. The results with the original cumulant model change according to the relaxation rate related to the kinematic viscosities, and the droplet is not circular at equilibrium state. The central moment forcing scheme is adopted to the improved cumulant model, and the non-physical phenomenon of noncircular droplets has been resolved. By changing the radius of the droplet, the value of the surface tension was obtained by satisfying Laplace's law. Finally, the problem of droplet impact on a liquid film is conducted using the densities of liquid and vapor, and the surface tension. The results are in good agreement with the analytical solution. Furthermore, the improved cumulant model provides more stable results even at low viscosities, unlike the RM and CM models. The proposed model can be straightforwardly extended to three-dimensional space as discussed in Appendix C. Our improved cumulant model is expected to be useful for phase change applications, where the stability is particularly important, along with the energy conservation equation.

Data Availability The data shown in this article are available from the corresponding author upon a reasonable request.

Declarations

Conflict of interest The authors declare no conflict of interest.

Appendix A Raw Moment Collision Model

In Appendix A, non-orthogonal raw moments, $m_{\alpha\beta}$, which collide in a fixed frame of reference, are considered. The raw moments, $m_{\alpha\beta}$, are defined by [3]

$$m_{\alpha\beta} = \frac{\partial^\alpha \partial^\beta}{\partial \Xi^\alpha \partial \Upsilon^\beta} \left(\sum_{i,j} f_{ij} e^{\Xi i} e^{\Upsilon j} \right) \Bigg|_{\Xi=\Upsilon=0} = \sum_{i,j} i^\alpha j^\beta f_{ij}. \tag{A.1}$$

The computation of raw moments can be simply split into two directions using a forward raw moment transformation [3]:

$$m_{i|\beta} = \sum_j f_{ij} j^\beta, \tag{A.2}$$

$$m_{\alpha\beta} = \sum_i m_{i|\beta} i^\alpha. \tag{A.3}$$

The RM forcing scheme can be expressed as

$$m^F_{\alpha\beta} = \sum_{i,j} i^\alpha j^\beta f^F_{ij}, \tag{A.4}$$

with Eq. (47), where $m^F_{\alpha\beta}$ are the discrete raw moment forcing terms.

Cumbersome calculations with Eqs. (A.4), (47) give the following expressions:

$$m^F_{00} = 0, \tag{A.5}$$

$$m_{10}^F = F_x, \tag{A.6}$$

$$m_{01}^F = F_y, \tag{A.7}$$

$$m_{20}^F = 2F_x u, \tag{A.8}$$

$$m_{02}^F = 2F_y v, \tag{A.9}$$

$$m_{11}^F = F_x v + F_y u, \tag{A.10}$$

$$m_{12}^F = \frac{F_x}{3}(1+3v^2) + 2F_y uv, \tag{A.11}$$

$$m_{21}^F = \frac{F_y}{3}(1+3u^2) + 2F_x uv, \tag{A.12}$$

$$m_{22}^F = \frac{2}{3}F_x u(1+3v^2) + \frac{2}{3}F_y v(1+3u^2) \tag{A.13}$$

The post-collision raw moments are obtained as:

$$m_{00}^* = \rho, \tag{A.14}$$

$$m_{10}^* = \rho u + \frac{m_{10}^F}{2}, \tag{A.15}$$

$$m_{01}^* = \rho v + \frac{m_{01}^F}{2}, \tag{A.16}$$

$$m_{20}^* - m_{02}^* = \omega_1 \rho \left(u^2 - v^2\right) + (1-\omega_1)(m_{20} - m_{02}) + \left(1 - \frac{\omega_1}{2}\right)(m_{20}^F - m_{02}^F), \tag{A.17}$$

$$m_{20}^* + m_{02}^* = \omega_2 \rho \left(u^2 + v^2 + \frac{2}{3}\right) + (1-\omega_2)(m_{20} + m_{02}) + \left(1 - \frac{\omega_2}{2}\right)(m_{20}^F + m_{02}^F), \tag{A.18}$$

$$m_{11}^* = \omega_1 \rho uv + (1-\omega_1)m_{11} + \left(1 - \frac{\omega_1}{2}\right) m_{11}^F, \tag{A.19}$$

$$m_{12}^* = \omega_3 \frac{\rho u}{3} + (1-\omega_3)m_{12} + \left(1 - \frac{\omega_3}{2}\right) m_{12}^F, \tag{A.20}$$

$$m_{21}^* = \omega_3 \frac{\rho v}{3} + (1-\omega_3)m_{21} + \left(1 - \frac{\omega_3}{2}\right) m_{21}^F, \tag{A.21}$$

$$m_{22}^* = \omega_4 \frac{\rho}{3}\left(u^2 + v^2 + \frac{1}{3}\right) + (1-\omega_4)m_{22} + \left(1 - \frac{\omega_4}{2}\right) m_{22}^F. \tag{A.22}$$

The post-collision raw moments are transformed back to the velocity space using a splitting method as found in [3].

Specifically, the backward raw moment transformation is as follows:

$$m_{0|\beta}^* = m_{0\beta}^* - m_{2\beta}^*, \tag{A.23}$$

$$m_{\bar{1}|\beta}^* = (-m_{1\beta}^* + m_{2\beta}^*)/2, \tag{A.24}$$

$$m_{1|\beta}^* = (m_{1\beta}^* + m_{2\beta}^*)/2, \tag{A.25}$$

$$f_{i0}^* = m_{i|0}^* - m_{i|2}^*, \tag{A.26}$$

$$f_{i\bar{1}}^* = (-m_{i|1}^* + m_{i|2}^*)/2, \tag{A.27}$$

$$f_{i1}^* = (m_{i|1}^* + m_{i|2}^*)/2. \tag{A.28}$$

Appendix B Central Moment Collision Model

A forward central moment transformation is applied to the particle distribution functions as Eqs. (20)-(21). Subsequently, the CM forcing scheme shown in Eqs. (49)-(52) is used for the collision process.

Collision:

$$k_{00}^* = \rho, \tag{B.1}$$

$$k_{10}^* = \frac{F_x}{2}, \tag{B.2}$$

$$k_{01}^* = \frac{F_y}{2}, \tag{B.3}$$

$$k_{20}^* - k_{02}^* = (1-\omega_1)(k_{20} - k_{02}), \tag{B.4}$$

$$k_{20}^* + k_{02}^* = \omega_2 \frac{2}{3}\rho + (1-\omega_2)(k_{20} + k_{02}), \tag{B.5}$$

$$k_{11}^* = (1-\omega_1)k_{11}, \tag{B.6}$$

$$k_{12}^* = (1-\omega_3)k_{12} + \left(1 - \frac{\omega_3}{2}\right)\frac{F_x}{3}, \tag{B.7}$$

$$k_{21}^* = (1-\omega_3)k_{21} + \left(1 - \frac{\omega_3}{2}\right)\frac{F_y}{3}, \tag{B.8}$$

$$k_{22}^* = \omega_4 \frac{\rho}{9} + (1-\omega_4)k_{22}. \tag{B.9}$$

After the collision process, a backward central moment transformation is applied to the post-collision central moments, similar to Eqs. (62)-(67).

Appendix C Improved Forcing Scheme for the Cumulant Collision Model in Three-Dimensional Space

Except for the part of the improved cumulant forcing scheme, the improved cumulant collision model in three-dimensional space is identical to the original cumulant collision model in [3]. Therefore, only the improved cumulant forcing scheme is shown in Appendix C.

The central moment forcing scheme can be extended to three-dimensional space given by [35]

$$k_{\alpha\beta\gamma}^F = \sum_{i,j,k} (i-u)^\alpha (j-v)^\beta (k-w)^\gamma f_{ijk}^F, \tag{C.1}$$

with

$$
\begin{aligned}
f_{ijk}^{F} = \omega_{ijk}\Bigg[& 3(iF_x + jF_y + kF_z) + \frac{3^2}{2}\Big\{\mathcal{H}_{xx}^{(2)}2uF_x + \mathcal{H}_{yy}^{(2)}2vF_y + \mathcal{H}_{zz}^{(2)}2wF_z \\
& + 2ij(uF_y + vF_x) + 2ik(uF_z + wF_x) + 2jk(vF_z + wF_y)\Big\} \\
& + \frac{3^3}{2}\Big\{\mathcal{H}_{xx}^{(2)}j(2uvF_x + u^2F_y) + \mathcal{H}_{xx}^{(2)}k(2uwF_x + u^2F_z) + i\mathcal{H}_{yy}^{(2)}(2uvF_y + v^2F_x) \\
& + i\mathcal{H}_{zz}^{(2)})(2uwF_z + w^2F_x) + j\mathcal{H}_{zz}^{(2)}(2vwF_z + w^2F_y) + \mathcal{H}_{yy}^{(2)}k(2vwF_y + v^2F_z) \\
& + 2ijk(uvF_z + uF_yw + F_xvw)\Big\} \\
& + \frac{3^4}{4}\Big\{\mathcal{H}_{xx}^{(2)}\mathcal{H}_{yy}^{(2)}(2u^2vF_y + 2uv^2F_x) + \mathcal{H}_{xx}^{(2)}\mathcal{H}_{zz}^{(2)}(2u^2wF_z + 2uw^2F_x) \\
& + \mathcal{H}_{yy}^{(2)}\mathcal{H}_{zz}^{(2)}(2v^2wF_z + 2vw^2F_y) + 2ij\mathcal{H}_{zz}^{(2)}(2uvwF_z + uF_yw^2 + F_xvw^2) \\
& + 2i\mathcal{H}_{yy}^{(2)}k(2uvwF_y + uv^2F_z + F_xv^2w) + 2\mathcal{H}_{xx}^{(2)}jk(2uvwF_x + u^2vF_z + u^2wF_y)\Big\} \\
& + \frac{3^5}{4}\Big\{\mathcal{H}_{xx}^{(2)}\mathcal{H}_{yy}^{(2)}k(2uv^2wF_x + 2u^2vwF_y + u^2v^2F_z) \\
& + \mathcal{H}_{xx}^{(2)}j\mathcal{H}_{zz}^{(2)}(2uvw^2F_x + u^2w^2F_y + 2u^2vwF_z) \\
& + i\mathcal{H}_{yy}^{(2)}\mathcal{H}_{zz}^{(2)}(v^2w^2F_x + 2uvw^2F_y + 2uv^2wF_z)\Big\} \\
& + \frac{3^6}{8}\Big\{\mathcal{H}_{xx}^{(2)}\mathcal{H}_{yy}^{(2)}\mathcal{H}_{zz}^{(2)}(2uv^2w^2F_x + 2u^2vw^2F_y + 2u^2v^2wF_z\Big\}\Bigg],
\end{aligned}
\tag{C.2}
$$

where w_{ijk} are the weight given by

$$
w_{ijk} = \frac{8}{27}\frac{1}{4^{|i|+|j|+|k|}}, \tag{C.3}
$$

and $\alpha, \beta, \gamma \in \{0, 1, 2\}$, $i, j, k \in \{1, 0, \bar{1}\}$, (u, v, w) are the fluid velocities for the (x, y, z) directions, and (F_x, F_y, F_z) are the forces in the (x, y, z) directions with the space variables (x, y, z). In Eq. (C.2), second-order Hermite polynomials such as $\mathcal{H}_{xx}^{(2)}$, $\mathcal{H}_{yy}^{(2)}$, and $\mathcal{H}_{zz}^{(2)}$ are given by [35]

$$
\mathcal{H}_{xx}^{(2)} = i^2 - 1/3, \tag{C.4}
$$

$$
\mathcal{H}_{yy}^{(2)} = j^2 - 1/3, \tag{C.5}
$$

$$
\mathcal{H}_{zz}^{(2)} = k^2 - 1/3. \tag{C.6}
$$

After cumbersome calculations with Eqs. (C.1)-(C.6), the following expressions are determined by

$$
k_{100}^{F} = F_x, \tag{C.7}
$$

$$
k_{010}^{F} = F_y, \tag{C.8}
$$

$$
k_{001}^{F} = F_z, \tag{C.9}
$$

$$
k_{120}^{F} = \frac{F_x}{3}, \tag{C.10}
$$

$$
k_{102}^{F} = \frac{F_x}{3}, \tag{C.11}
$$

$$
k_{210}^{F} = \frac{F_y}{3}, \tag{C.12}
$$

$$k^F_{201} = \frac{F_z}{3}, \tag{C.13}$$

$$k^F_{012} = \frac{F_y}{3}, \tag{C.14}$$

$$k^F_{021} = \frac{F_z}{3}, \tag{C.15}$$

$$k^F_{122} = \frac{F_x}{9}, \tag{C.16}$$

$$k^F_{212} = \frac{F_y}{9}, \tag{C.17}$$

$$k^F_{221} = \frac{F_z}{9}, \tag{C.18}$$

where all other values of $k^F_{\alpha\beta\gamma}$ are zero.

The improved cumulant forcing scheme using Eqs. (C.7)-(C.18):

$$\tilde{k}^*_{000} = \rho, \tag{C.19}$$

$$\tilde{k}^*_{100} = \frac{F_x}{2}, \tag{C.20}$$

$$\tilde{k}^*_{010} = \frac{F_y}{2}, \tag{C.21}$$

$$\tilde{k}^*_{001} = \frac{F_z}{2}, \tag{C.22}$$

$$\tilde{k}^*_{110} = k^*_{110}, \tag{C.23}$$

$$\tilde{k}^*_{101} = k^*_{101}, \tag{C.24}$$

$$\tilde{k}^*_{011} = k^*_{011}, \tag{C.25}$$

$$\tilde{k}^*_{200} - \tilde{k}^*_{020} = k^*_{200} - k^*_{020}, \tag{C.26}$$

$$\tilde{k}^*_{200} - \tilde{k}^*_{002} = k^*_{200} - k^*_{002}, \tag{C.27}$$

$$\tilde{k}^*_{200} + \tilde{k}^*_{020} + \tilde{k}^*_{002} = k^*_{200} + k^*_{020} + k^*_{002}, \tag{C.28}$$

$$\tilde{k}^*_{120} + \tilde{k}^*_{102} = k^*_{120} + k^*_{102} + \left(1 - \frac{\omega_3}{2}\right)\frac{2F_x}{3}, \tag{C.29}$$

$$\tilde{k}^*_{210} + \tilde{k}^*_{012} = k^*_{210} + k^*_{012} + \left(1 - \frac{\omega_3}{2}\right)\frac{2F_y}{3}, \tag{C.30}$$

$$\tilde{k}^*_{201} + \tilde{k}^*_{021} = k^*_{201} + k^*_{021} + \left(1 - \frac{\omega_3}{2}\right)\frac{2F_z}{3}, \tag{C.31}$$

$$\tilde{k}^*_{120} - \tilde{k}^*_{102} = k^*_{120} - k^*_{102}, \tag{C.32}$$

$$\tilde{k}^*_{210} - \tilde{k}^*_{012} = k^*_{210} - k^*_{012}, \tag{C.33}$$

$$\tilde{k}^*_{201} - \tilde{k}^*_{021} = k^*_{201} - k^*_{021}, \tag{C.34}$$

$$\tilde{k}^*_{111} = k^*_{111}, \tag{C.35}$$

$$\tilde{k}^*_{220} - 2\tilde{k}^*_{202} + \tilde{k}^*_{022} = k^*_{220} - 2k^*_{202} + k^*_{022}, \tag{C.36}$$

$$\tilde{k}^*_{220} + \tilde{k}^*_{202} - 2\tilde{k}^*_{022} = k^*_{220} + k^*_{202} - 2k^*_{022}, \tag{C.37}$$

$$\tilde{k}^*_{220} + \tilde{k}^*_{202} + \tilde{k}^*_{022} = k^*_{220} + k^*_{202} + k^*_{022}, \tag{C.38}$$

$$\tilde{k}^*_{211} = k^*_{211}, \tag{C.39}$$

$$\tilde{k}^*_{121} = k^*_{121}, \tag{C.40}$$

$$\tilde{k}^*_{112} = k^*_{112}, \tag{C.41}$$

$$\tilde{k}^*_{122} = k^*_{122} + \left(1 - \frac{\omega_9}{2}\right)\frac{F_x}{9}, \tag{C.42}$$

$$\tilde{k}^*_{212} = k^*_{212} + \left(1 - \frac{\omega_9}{2}\right)\frac{F_y}{9}, \tag{C.43}$$

$$\tilde{k}^*_{221} = k^*_{221} + \left(1 - \frac{\omega_9}{2}\right)\frac{F_z}{9}, \tag{C.44}$$

$$\tilde{k}^*_{222} = k^*_{222}, \tag{C.45}$$

where ω_3 and ω_9 are free parameters with a value between 0 and 2.

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